



Bringing Innovation to Life Through Immersive AR/VR

Background:

- Ahmedabad Textile Industry's Research Association (ATIRA), a leading textile research institute with over seven decades of contribution, has played a key role in indigenous innovation, applied research, and industry capacity building. One of the key research is development of **nanofiber-based indigenous HEPA filters** aimed at improving air quality across industrial and commercial applications.
- During a visit to the AI Centre of Excellence (AI CoE), ATIRA explored ways to enhance outreach and communicate this science and its application to a broad audience. At AI CoE's Experience Zone, ATIRA's leaders interacted with an immersive AR/VR solution made by a startup named Exposit.

Solution:

- The AI CoE enabled structured collaboration between ATIRA and Exposit to co-create this solution. The resulting VR experience presents the full HEPA lifecycle —covering concept, material science, nanofiber filtration, testing, validation, and real-world applications.

Impact:

- ATIRA Demonstrated the solution at textile conference where 300+ people experienced the complex science of building nanofiber based HEPA filters in minutes, without prior technical background
- The Solution is permanently housed at ATIRA as a knowledge asset for continuous industry engagement



I personally see immersive technologies as a powerful enabler for bridging the gap between research and real-world understanding. The AR/VR use case facilitated by the AI CoE has transformed a complex R&D journey into an intuitive and impactful experience for industry, academia, and policymakers alike. The support in onboarding the right technology partner has been pivotal in delivering this outcome.

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